

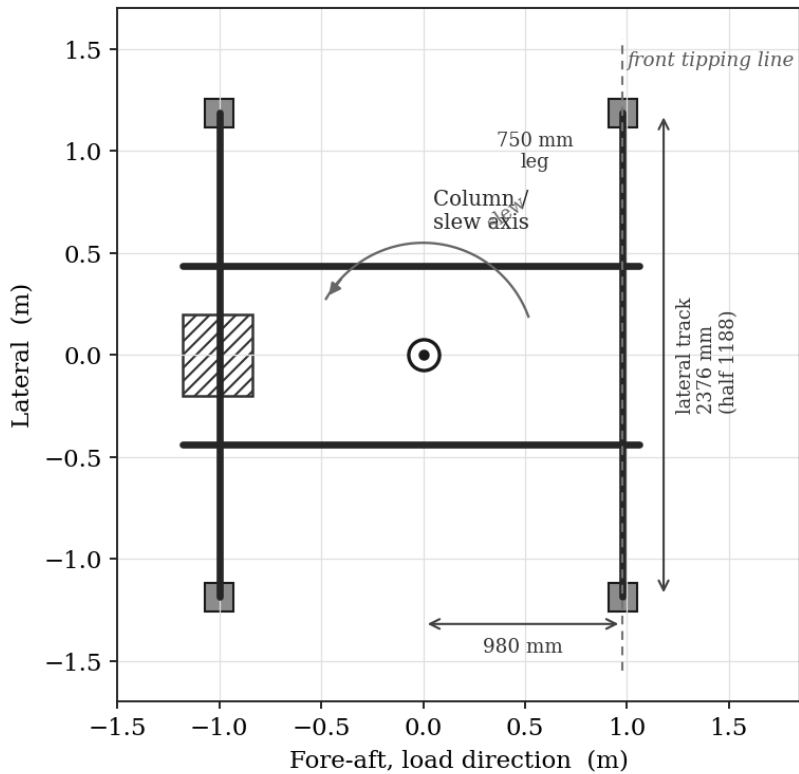
PROJECT 40467361

STRUCTURAL CALCULATION AND
LOAD VERIFICATION REPORT

1 Tonne Mobile Floor Jib Crane

Deployed-outrigger configuration: stability in the # arrangement and the leg-member check

Deployed footprint, plan view: two rails plus four transverse legs (# arrangement)



Deployed footprint with the four legs in the # arrangement. The legs stand square to the rails, so they widen the lateral track while the front line stays 980 mm ahead of the column.

Field	Detail
Project reference	40467361
Configuration	Outriggers deployed (four transverse legs, # arrangement)
Rated capacity (as marked)	1000 kg at 1400 mm reach
Document revision	Rev B, supersedes Rev A on the corrected front-overhang geometry
Date	11 June 2026
Status	Issued for review
Analysis tool	MATLAB R2026a, parametric script C2_v2

Document Control

This report assesses the 1 tonne mobile jib crane on the manufacturer drawing set with its four outrigger legs deployed. It is the companion to the wheels-only Rev 2 report, which covers the same crane standing on its wheels with the legs stowed. Both documents run off one parametric model and one input set. This revision, Rev B, replaces the earlier Rev A: the front overhang has been corrected from the 1200 mm place-holder to 980 mm, now that the full base drawing and the warning plate together fix that dimension. The change makes forward stability worse, so the headline result moves with it.

Revision history

Rev	Date	Description	Status
A	10 Jun 2026	First issue of the deployed-outrigger study, with the front line taken at the 1200 mm place-holder.	Superseded
B	11 Jun 2026	Front overhang corrected to 980 mm from the base top-view chain and the warning-plate 420 mm dimension. Forward-stability and leg-reaction figures revised. Items above the slewing bearing carried from Rev 2.	Issued for review

Basis and limitations

- Drawings as issued. Leg dimensions come from Base DETAIL-11, DETAIL B-B and DETAIL-12. Section sizes, the frame track and the front overhang come from the base and assembly sheets.
- Front overhang resolved. The 980 mm front line follows from two readings that agree: the base top-view chain places the slew centre about 1076 mm behind the front edge, and the warning plate marks 420 mm at the front, which puts the 1400 mm load 420 mm beyond the front line, i.e. the line sits at 1400 minus 420, or 980 mm.
- Deployed geometry. The legs are taken square to the rails at 90 degrees, as the deployment schematic shows. The deployed foot positions are part-dimensioned on the drawings and are flagged in Section 11.
- Scope of physics. Static lifting only. Stability is a moment balance on static loads. Wind, side pull and snatch are outside this scope, as in Rev 2.
- Carried-over results. The column, boom, pins, welds and slewing bolts sit above the slewing bearing and do not change with the legs. Their results come from Rev 2 and are summarised here, not re-derived. The base outrigger arm and the wheel and foot reactions do shift slightly, because the front line moved, and those are reported with their corrected values.
- Not a design. This verifies an existing unit against its rated chart in the deployed state. It does not replace load testing or a competent-person examination before the crane is put to work.

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1. Executive Summary

This report checks the 1 tonne mobile jib crane covered by the wheels-only Rev 2 report, but with the four outrigger legs deployed. The legs form a # arrangement. The two fore-aft rails that carry the wheels run front to back, and a front and a rear transverse beam cross them, each reaching out to either side to a foot. Seen from above, the rails and the two leg beams read as a #.

One feature of that layout decides the result. The legs sit at 90 degrees to the rails, so they reach sideways and not forward. Putting them down is what lets the crane slew, because the wide lateral track keeps it from going over when the boom swings off the fore-aft line. The legs do nothing for forward tipping, because a transverse leg adds no reach in the luffing plane. The front tipping line stays at the front frame line, 980 mm ahead of the column, the same place it sits on the wheels.

Forward stability is therefore identical to the wheels-only case, and on the corrected 980 mm geometry that case is worse than Rev A reported. The mid-reach point, 700 kg at 2320 mm, governs, and its factor against tipping is 0.96. A factor below one means the lifted-load moment about the front line already exceeds the restoring moment, so on a static balance the crane is past its forward tipping point at that chart position. The earlier 1.34 came from the optimistic 1200 mm front line and does not hold.

Sideways stability is the question the legs were meant to answer, and it is unchanged from Rev A because it does not depend on the front line. With the 2376 mm track that the 750 mm legs give, the crane is steady close in, at a factor of 3.92 at 1400 mm, but it tips at medium and long reach, where the factors fall to 1.05 and 1.31. Reaching 1.50 at the worst point needs the track opened to about 2784 mm. The manufacturer figure shows the feet adjusting wider, toward 2870 mm, so the deployed track has to be measured on the machine.

The deployed leg member is sound. Taken as a 750 mm cantilever in RHS 60 x 100 x 6 carrying half the front reaction, it reaches 144 MPa for a factor of 1.91. The base outrigger arm, which Rev A reported as failing at 1.38, now clears the target at 1.53, because the shorter front overhang shortens its cantilever. Both front-support members pass.

Nothing above the slewing bearing changes. The legs bolt to the base frame, below the bearing, so the axial load and the moment that travel up into the column come from the boom and the lifted load alone. The column stays at yield, factor 0.99, its base weld stays below target at 0.85, and the boom outer section stays marginal at 1.48, the same figures the wheels-only report gives. Deploying the legs rescues none of these.

The conclusion follows directly. The legs make slewing possible and they ease the front-support members, but forward tipping is unchanged and now sits below one at mid-reach, sideways stability is short until the track is widened and confirmed, and the column and weld limits are untouched. The crane cannot be signed off to the full 1 tonne chart on a working-stress basis. The route to the structural items, the re-rate or the column upgrade, is the one set out in the Rev 2 report and it does not depend on the legs.

Headline results, deployed configuration (# arrangement, legs at 90 degrees)

Item	Result	FoS	Target	Verdict
Forward tipping, worst (700 kg @ 2320 mm)	9.20 kN·m	0.96	1.50	Fail (tips)
Sideways tipping, worst (700 kg @ 2320 mm)	7.77 kN·m	1.05	1.50	Fail

Item	Result	FoS	Target	Verdict
Deployed leg member (RHS 60x100x6)	144 MPa	1.91	1.50	Pass
Base outrigger arm (corrected geometry)	180 MPa	1.53	1.50	Pass
Column, combined N+M (carried over)	277 MPa	0.99	1.50	Fail
Column-to-base weld (carried over)	235 MPa	0.85	1.50	Fail
Lateral track provided	2376 mm	—	2784 mm	Short

Section 9 places the deployed and wheels-only results side by side. Sections 5, 6 and 7 give the stability balances and the leg-member calculation in full.

2. Introduction and Scope

2.1 Background

The unit is a mobile floor jib crane rated to lift 1 tonne. It runs on four wheels, carries a counterweight at the rear, and luffs a three-stage telescopic boom on a single hydraulic cylinder. The upper assembly slews on a bolted bearing flange above the base frame. The warning plate fitted to the machine tells the operator to put the legs down before slewing, so the rated chart is written for the legs-down state.

The wheels-only Rev 2 report covers the crane standing on its wheels with the legs stowed. This report is the companion study for the deployed state. It looks at what the four legs change and at what they leave alone.

2.2 What this report covers

The deployed legs act on the part of the crane below the slewing bearing, which is stability and the members that carry the load down to the ground. This report covers:

- Forward tipping in the luffing plane, taken about the front foot line
- Sideways tipping with the boom slewed across the rails, taken about a side foot line
- The track the legs need for sideways stability to reach the target, against the track they give
- Bending in the deployed leg member, and the base outrigger arm it shares load with
- A statement of which wheels-only results carry over without change, and why

Everything above the slewing bearing, the column, the boom, the pins and the slewing bolts, is fixed by the boom and the lifted load and does not change with the legs. Those checks come from Rev 2 and are summarised in Section 8 rather than worked again.

2.3 Configuration assessed

Deployed configuration

All stability results in this report are for the crane standing on its four deployed legs in the # arrangement, with the legs square to the rails at 90 degrees as the deployment schematic shows. The wheels-only results appear only for comparison. If the legs in fact splay forward, forward stability will be better than reported here; Section 11 flags this.

3. The Deployed Configuration

The crane base is a pair of fore-aft rails set about 876 mm apart, with a wheel at each corner. Stowed, that narrow rail track is the whole footprint, and it is the reason the crane is unsafe to slew on its wheels. Deploying the legs adds two transverse beams, one on the front wheel line and one toward the rear, each reaching out to both sides. The four feet at their ends, with the two rails, make the # seen in plan.

The leg dimensions come from Base DETAIL-11. Each leg arm is 750 mm long, with a 230 mm drop to an 80 mm foot pad, an M16 jack screw for levelling, and a Ø32 hinge pin. DETAIL B-B records a longer 798 mm arm and DETAIL-12 a Ø20 eye foot. The 750 mm arm of DETAIL-11 is taken as the deployed reach. Because the beams sit square to the rails, the feet widen the footprint sideways only. The front foot stays level with the front frame line, 980 mm ahead of the column, so it adds nothing to the forward overhang.

The lateral track follows from the geometry. Each foot sits half the frame track plus one leg arm out from the centreline, so the half-track is 438 plus 750, which is 1188 mm, and the full track is 2376 mm. This single number governs sideways stability and is carried through Section 6.

Quantity	Value	Source
Leg arm (lateral reach)	750 mm	DETAIL-11
Foot drop	230 mm	DETAIL-11
Foot pad	80 mm	DETAIL-11
Levelling jack screw	M16 (Ø16)	DETAIL-11
Leg hinge pin	Ø32	DETAIL-11
Alternative leg arm	798 mm	DETAIL B-B
Foot eye	Ø20	DETAIL-12
Frame track, rail to rail	876 mm	Base drawing
Deployed lateral track	2376 mm (876 + 2 × 750)	derived
Front foot ahead of column	980 mm (front frame line)	Base drawing + warning plate
Leg member section	RHS 60 x 100 x 6	drawing

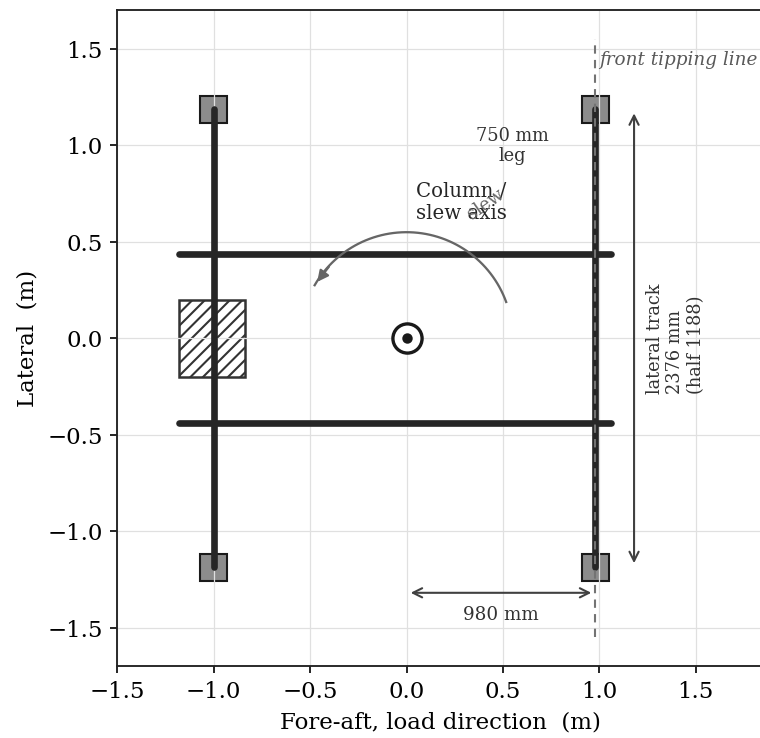
Deployed footprint, plan view: two rails plus four transverse legs (# arrangement)

Figure 3.1 Deployed footprint in plan. The legs reach 750 mm to each side, giving a 2376 mm track, while the front foot stays on the 980 mm front line.

4. Materials and Load Model

Materials, sections and the load model are unchanged from Rev 2 and are repeated here only so the report stands on its own. All members are hollow steel sections in grade S275, with a yield of 275 MPa. The member allowable is the yield divided by the 1.5 target. Pins carry a 3.0 target, as is normal for lifting connections.

Allowable stresses (as Rev 2)

Member bending / direct stress (target FoS 1.5):

$$\sigma_{\text{allow}} = f_y / 1.50 = 275 / 1.50 = 183.3 \text{ MPa}$$

Pin / bolt shear (target FoS 3.0):

$$\tau_{\text{allow}} = 0.6 \times f_y / 3.0 = 55.0 \text{ MPa}$$

The rated chart gives three points, with capacity falling as reach grows. The factored load applies the 1.25 dynamic factor used for HC1 hoist duty.

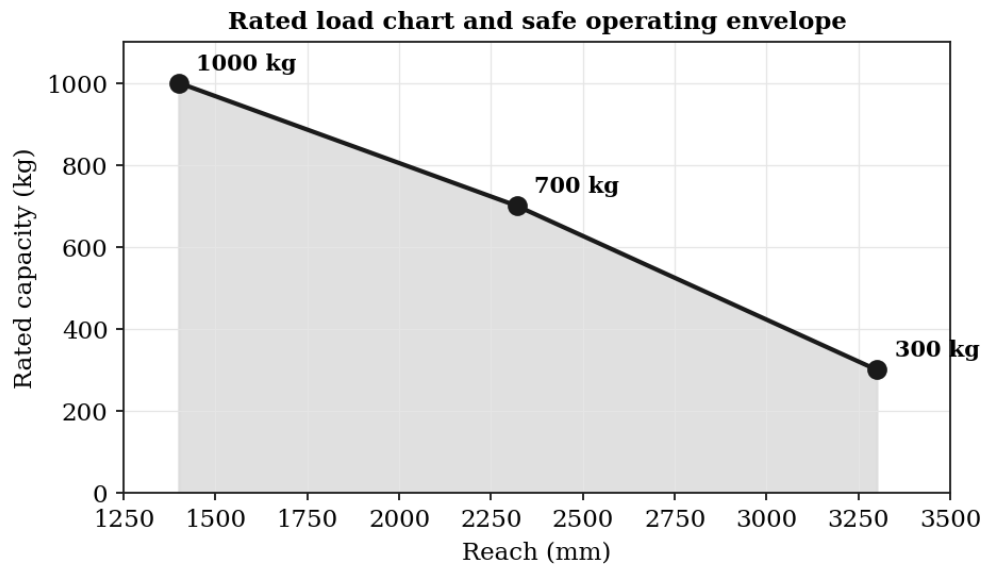


Figure 4.1 Rated load chart with the safe operating envelope shaded.

Chart point	Reach (mm)	Rated load (kg)	Static load (kN)	Factored load (kN)
Close	1400	1000	9.81	12.26
Mid	2320	700	6.87	8.58
Far	3300	300	2.94	3.68

The structural items carried over in Section 8 are worst at the mid-reach point, where the design moment at the column is 19.91 kN·m. Stability, treated next, is assessed on the static loads, because the 1.25 dynamic factor applies equally to the overturning and the restoring sides and cancels in the ratio.

5. Forward Stability

Forward tipping is a moment balance about the front foot line, 980 mm ahead of the column. With the legs at 90 degrees, that line sits in the same place as the front wheel, so the balance is identical to the wheels-only case. The restoring moment comes from the crane weight and counterweight behind the line, and the overturning moment from the lifted load ahead of it.

Overturning balance about the front foot line

$$\begin{aligned}\text{Restoring } M_R &= m_{\text{crane}} \times g \times (L_{\text{fwd}} + e_{\text{cg}}) \\ &= 700 \times 9.81 \times (0.980 + 0.300) = 8.79 \text{ kN.m}\end{aligned}$$

$$\text{Overturning } M_{OT} = m_{\text{load}} \times g \times (\text{reach} - L_{\text{fwd}}) \quad [\text{static load}]$$

Load case	M_OT (kN·m)	FoS = M_R / M_OT	Verdict
1000 kg @ 1400 mm	4.12	2.13	Pass
700 kg @ 2320 mm	9.20	0.96	Fail (tips)
300 kg @ 3300 mm	6.83	1.29	Fail

The mid-reach point governs at a factor of 0.96, below one and well below the 1.50 target. A factor under one means the overturning moment already beats the restoring moment, so on the static balance the crane is over its forward tipping point at that position. This is the figure the corrected 980 mm front line produces, and it is worse than the 1.34 Rev A reported on the 1200 mm place-holder. Transverse legs do not move the front line, so they cannot improve forward stability.

6. Sideways Stability During Slew

Sideways tipping is the case the legs are there for. With the boom slewed across the rails, the crane can go over a side foot line set at the half-track, 1188 mm from the centreline. The restoring moment is the crane weight acting at that half-track. The overturning moment is the lifted load acting at its reach, less the half-track, since the foot line is offset toward the load by that amount. This balance does not involve the front line, so it is unchanged from Rev A.

Overturning balance about a side foot line

Restoring $M_R = m_{\text{crane}} \times g \times t_{\text{half}} = 700 \times 9.81 \times 1.188 = 8.16 \text{ kN.m}$

Overturning $M_{OT} = m_{\text{load}} \times g \times (\text{reach} - t_{\text{half}})$ [static load]

Half-track from the legs: $t_{\text{half}} = 438 + 750 = 1188 \text{ mm}$ (track 2376 mm)

Load case	$M_{OT} \text{ (kN.m)}$	$\text{FoS} = M_R / M_{OT}$	Verdict
1000 kg @ 1400 mm	2.08	3.92	Pass
700 kg @ 2320 mm	7.77	1.05	Fail
300 kg @ 3300 mm	6.22	1.31	Fail

The close-in lift is steady, but the mid and far points both fall below target, at 1.05 and 1.31. The mid-reach point is again the worst, and it sets how wide the feet must stand. Solving the balance for a factor of 1.50 at that point gives a half-track of 1392 mm, a full track of 2784 mm. The legs as drawn give 2376 mm, about 400 mm short.

The shortfall is in the track, not in the legs themselves. The manufacturer figure marks the deployed width as adjustable up to roughly 2870 mm, and at that full width the worst sideways factor lifts to about 1.6. The deployed foot positions therefore have to be measured on the machine and set to at least 2784 mm before the crane is slewed at mid or long reach.

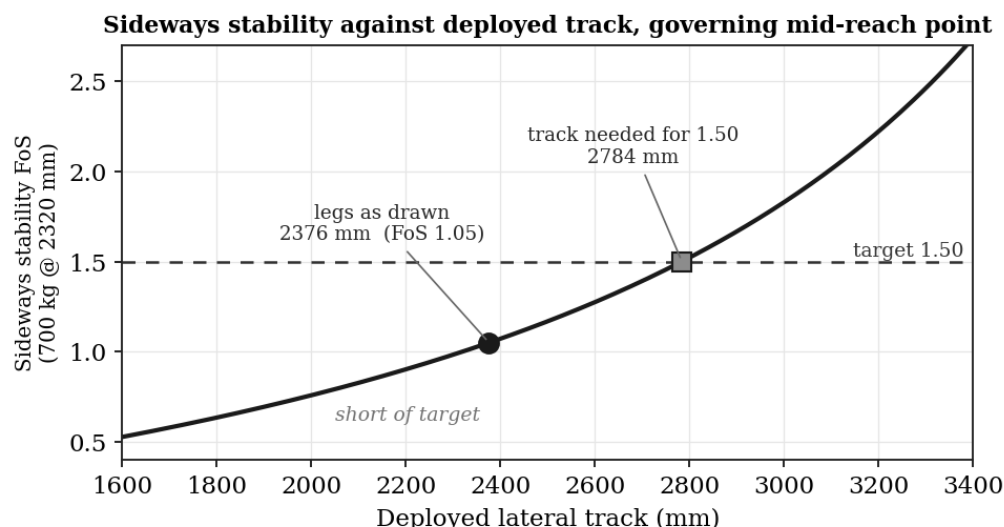


Figure 6.1 Sideways stability against the deployed track at the governing mid-reach point. The 2376 mm track from the legs sits below the 1.50 target; about 2784 mm is needed.

7. The Deployed Leg Member

The deployed leg is checked as a cantilever picking up the vertical reaction at its foot. The largest foot reaction comes from the heaviest lift taken in close, not from the tipping case, because member force follows the vertical load while tipping follows the lever arm. On the corrected geometry the front pair carries 17.43 kN at 1000 kg and 1400 mm, so each leg arm carries about 8.72 kN at its 750 mm reach.

Deployed leg, RHS 60 x 100 x 6, cantilever

front reaction (1000 kg @ 1400 mm) = 17.43 kN total -> 8.72 kN per leg

lever (leg arm) = 750 mm = 0.750 m

$M_{leg} = 8.72 \times 0.750 = 6.54 \text{ kN.m}$

$S_x = 45.5 \text{ cm}^3 = 45.5 \times 10^{-6} \text{ m}^3$

$\sigma = M_{leg} / S_x = 6.54 \times 10^6 / 45.5 \times 10^{-6} = 143.7 \text{ MPa}$

$FoS = f_y / \sigma = 275 / 143.7 = 1.91 \rightarrow \text{Pass}$

The leg passes at 1.91. The base outrigger arm, which carries the same front reaction over its fore-aft cantilever to the front wheel line, comes out at 180 MPa for a factor of 1.53 on the corrected geometry. Rev A reported that member as failing at 1.38, on the longer 1200 mm cantilever; the shorter 980 mm front line brings it over the target. Both front-support members are now sound.

8. Items Carried Over from the Wheels-Only Check

The legs attach to the base frame, below the slewing bearing. The axial load and the bending moment that pass up through the bearing into the column, and on into the boom and its pins, are fixed by the boom geometry and the lifted load. None of them change when the base stands on legs instead of wheels. The wheels-only results for those members therefore apply without alteration. The base outrigger arm, which does sit below the bearing and does move with the front line, is shown with its corrected value. The set is collected below.

Check	FoS	Target	Verdict
Boom outer (100x140x6)	1.48	1.50	Marginal
Boom middle (80x120x6)	2.54	1.50	Pass
Boom inner (60x100x6)	3.09	1.50	Pass
Column, combined N+M	0.99	1.50	Fail
Column buckling	67.6	1.50	Pass
Base outrigger arm (corrected)	1.53	1.50	Pass
Pivot pin (Ø30)	2.44	3.00	Fail
Cylinder rod-end pin (Ø20 as modelled)	1.20	3.00	Fail
Hook pin (Ø20)	4.03	3.00	Pass
Slewing bolts (5 x M16, 8.8)	1.69	1.50	Pass
Weld, column base	0.85	1.50	Fail
Weld, boom bracket	3.39	1.50	Pass

The cylinder rod-end pin is listed at the conservative 20 mm diameter and fails on that basis. As Rev 2 sets out, the only hinge pin shown near the cylinder mount is the Ø40 Base DETAIL-4 pin, which clears the check at 4.81; the as-built pin diameter should be confirmed on the machine. The column at 0.99 and its base weld at 0.85 remain the governing structural items in both configurations.

9. Stowed versus Deployed

The table below sets the two configurations side by side. The legs change the bottom of the load path and the stability picture. They leave the top of the structure exactly as it was.

Item	Wheels stowed	Legs deployed (# arrangement)
Front tipping line ahead of column	980 mm	980 mm (no forward reach)
Lateral foot track	876 mm	2376 mm
Forward tipping, worst point	FoS 0.96, Fail	FoS 0.96, Fail (unchanged)
Sideways tipping during slew	narrow track, tips	FoS 1.05, Fail (track short)
Front ground-support member	outrigger arm 180 MPa, FoS 1.53, Pass	leg arm 144 MPa, FoS 1.91, Pass
Column, combined N+M	277 MPa, FoS 0.99, Fail	unchanged
Column-to-base weld	235 MPa, FoS 0.85, Fail	unchanged
Boom outer section	186 MPa, FoS 1.48, Marginal	unchanged
Pins and slewing bolts	as Rev 2	unchanged

Read across, the pattern is consistent. The legs widen the track and ease the front-support members, so sideways tipping becomes possible to manage and both the leg and the outrigger arm pass. Forward tipping and the column, weld and boom are the same in both states, because the legs do not act on them.

Stability with the legs deployed (# arrangement, 90 deg transverse)

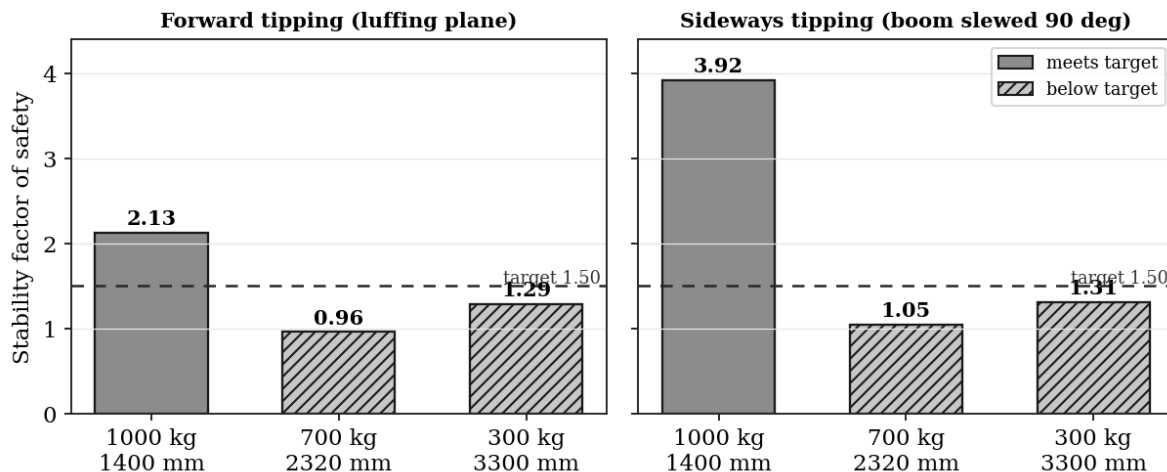


Figure 9.1 Stability in the deployed state on the corrected geometry. Forward tipping is unchanged from the wheels-only case and sits below one at mid-reach; sideways tipping clears the target only close in.

10. Conclusions and Recommendations

10.1 Conclusion on the deployed configuration

With the legs deployed in the # arrangement, the crane can be slewed, which it cannot do safely on its wheels. That is the real gain from putting the legs down. It does not make the crane compliant with its rated chart.

Forward tipping is unchanged from the wheels-only case and, on the corrected 980 mm front line, fails at mid-reach with a factor of 0.96, below one. Transverse legs add no forward reach, so they cannot help. Sideways tipping is governed by the deployed track. The 2376 mm track from the 750 mm legs clears the target only close in and falls short at mid and far reach, where about 2784 mm is needed. The deployed leg member is sound at 1.91, and the base outrigger arm now passes at 1.53. Everything above the slewing bearing is as the wheels-only report found it, with the column at yield, factor 0.99, and its base weld below target at 0.85.

So the crane cannot be certified to the full 1 tonne chart on a working-stress basis in the deployed configuration any more than on its wheels. The legs take care of sideways stability, provided the track is set wide enough, but they do not touch the forward-tipping shortfall or the structural items.

10.2 The governing items in the deployed state

- **Forward tipping, factor 0.96.** Now the leading stability item, below one at mid-reach on the corrected geometry. Carried over from the wheels-only case and not improved by transverse legs. Resolved only by re-rating the chart or cutting reach.
- **Sideways track, 2376 mm against 2784 mm needed.** The one item the deployed configuration itself must fix. Confirm the deployed width and, if the feet adjust, set the track to at least 2784 mm.
- **Column, factor 0.99, and column-base weld, factor 0.85.** Carried over and unchanged. The structural headline, addressed by the Rev 2 re-rate or upgrade.

10.3 Recommendations

Confirm the deployed footprint first. Measure the front foot position and the lateral track on the machine. If the legs angle forward at all, forward stability will be better than this report shows, and if the track opens past 2784 mm, sideways stability clears the target. Both turn on numbers the drawings give only in part.

Treat the structural and forward-tipping items through the Rev 2 report. Re-rating the chart to what the steel and the stability safely carry, near 400 to 450 kg at mid and far reach, clears every working-stress check in either configuration and needs no fabrication. Holding the full 1 tonne rating needs the larger column section and matching base weld set out there. Deploying the legs is a precondition for slewing, not a substitute for that work.

Recommendation

Do not slew the crane until the deployed track is confirmed to at least 2784 mm. Do not work the crane to the current chart until forward tipping, the column, the base weld and the cylinder pin are addressed per the wheels-only report. Putting the legs down is required before any slew, but on its own it does not make the rated chart safe.

11. Assumptions and Items to Confirm

The leg parts and the section sizes come straight off the drawings and are not in doubt. The deployed positions, and a few inputs shared with the wheels-only report, are estimates or part-dimensioned and should be confirmed against the built machine before the report is relied on.

- **Front foot ahead of column, 980 mm.** Resolved in this revision from the base top-view chain and the warning-plate 420 mm dimension, which agree. This replaces the 1200 mm place-holder used in Rev A and is the change that drives the revised forward-stability result. Worth a tape check on the machine, but the two independent readings give confidence.
- **Deployed lateral track, 2376 mm.** Derived from the 876 mm frame track and the 750 mm DETAIL-11 leg arm. This is the make-or-break input for sideways stability. The manufacturer figure suggests it can be set up to about 2870 mm; the as-built deployed width must be measured.
- **Leg deployment angle, 90 degrees.** Taken from the deployment schematic, with the legs square to the rails. The product photograph appears to show the front legs splayed forward. A forward splay would advance the front line and improve forward stability, which a square leg does not. The angle should be confirmed, since it drives both the forward foot position and the track.
- **Leg arm, 750 mm, and section, RHS 60 x 100 x 6.** From DETAIL-11 and the drawing. DETAIL B-B shows a 798 mm variant; confirm which is fitted.
- **Crane self-weight, 700 kg, and CG offset, 0.30 m behind the column.** Client-supplied and estimated respectively, as in Rev 2. A swing of plus or minus 100 kg moves both stability results, and the CG offset is the one input the drawings do not give, so it should be settled by weighing.
- **Carry-over items.** The column, weld, boom, pin and bolt results come from Rev 2 and inherit its assumptions, including nominal S275 yield and the 6 mm weld throat. See Rev 2 Section 14.

12. Standards Referenced

- EN 13001-3-1, crane structural design, load effects and proof of competence
- EN 1993-1-1, design of steel structures, general rules
- EN 1993-1-8, design of joints
- Cross-referenced with AS 1418.1 and SS 559 for the Singapore context

Appendix A: Companion files

- MATLAB R2026a parametric script (jib_crane_verification_C2_v2.m), the file driving this deployed study, with the front overhang set to 980 mm
- Console log (console_log.txt), the full run including the deployed stability tables and the leg-member check
- Wheels-only report (Rev 2) and the combined verification (Rev 3), the parent documents for the carried-over results
- Figures, grayscale, as used in this report